

- ⑩ parse tree
- ⑪ PDA, NPDA
- ⑫ PDA ex.

ML numericals

Section 1: Linear Regression

③ Gradient Descent
 ↳ SGD
 ↳ BGD

② Matrix Formulation
 ↳ Normal method
 ↳ multiple linear Regn
 ↳ polynomial (quadratic)

① Basic form
 ↳ calculus method
 ↳ peckhazur / least squares

① $y = \beta_0 + \beta_1 x$
 response coefficient predictor

Aim: To Train model & get best coefficients

→ input → $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ → $\hat{y}$ → cap / predicted
 ↳ predicted coefficients

⇒ Note: β_0, β_1 (or) β (set of coeffⁿ) need to be OPTIMIZED!!

⇒ while optimizing, minimize error $e_i = \hat{y}_i - y_i$

$$MSE = \frac{1}{n} \sum_{i=1}^n e_i^2 \quad : n = \text{number of observation}$$

⇒ Calculus method: (Direct formula + Derivation)

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_i, \text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad \text{RSS}$$

\$\Rightarrow \therefore \beta_0\$ & \$\beta_1\$ need to be changed for optimizing MSE,
differentiate MSE w.r.t. Both

$$\Rightarrow \frac{\partial (\text{MSE})}{\partial \beta_0} = \frac{2}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \xrightarrow{(1)} = 0 \text{ for minima}$$

$$\frac{\partial (\text{MSE})}{\partial \beta_1} = \frac{2}{n} \left(\sum_{i=1}^n (\hat{y}_i - y_i) x_i \right) \xrightarrow{(2)} = 0 \quad \left\{ \text{ye } \frac{\partial (\hat{y}_i)}{\partial \beta_1} \text{ aa rha hai!!} \right\}$$

\$\rightarrow\$ now we have 2 var. \$(\hat{\beta}_0, \hat{\beta}_1)\$

& 2 eqn (1) & (2)

$$\Rightarrow \text{eqn (1)} \Rightarrow \sum (\hat{y}_i - y_i) = 0$$

$$\sum \hat{y}_i - \sum y_i = 0 \quad (\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i)$$

$$\begin{cases} \Rightarrow \sum \hat{\beta}_0 + \sum \hat{\beta}_1 x_i = \sum y_i \\ \Rightarrow \frac{1}{n} \sum \hat{\beta}_0 + \hat{\beta}_1 \sum x_i = \sum y_i \quad \text{--- (1)'} \end{cases}$$

$$\approx \text{eqn (2)'} = \hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2 = \sum x_i y_i \quad \text{--- (2)'}$$

$$\xrightarrow{\text{solving}} \hat{\beta}_0 = \frac{\sum \hat{y}_i - \hat{\beta}_1 \sum x_i}{n} = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{n \sum x_i \hat{y}_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)(\sum x_i)} \Rightarrow \text{derivation done}$$

\$\Rightarrow\$ Least Squares method:
$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

given

x	y	$x - \bar{x}$	$y - \bar{y}$
1	1	-2	-1.8
2	3	-1	0.2
3	3	0	0.2
4	2	1	-0.8
5	5	2	2.2
$\bar{x} = 3$	$\bar{y} = 2.8$	$\sum^2 = (4 + 1 + 1 + 1 + 4) = 10$	

$$\Rightarrow \hat{\beta}_1 = \frac{7}{10} = 0.7 \quad (2)$$

$$\hat{\beta}_0 = 2.8 - 3 \times 0.7 = 0.7$$

$$\Rightarrow \text{equation} \Rightarrow \hat{y} = 0.7 + 0.7x$$

$$(x - \bar{x})(y - \bar{y})$$

$$\begin{array}{r} 3.6 \\ -0.2 \\ 0 \\ -0.8 \\ 4.4 \\ \hline \sum = 7 \end{array}$$

\$\Rightarrow\$ Reporting error

x	y
1	0.7 + 0.7(1) = 1.4
2	0.7 + 0.7(2) = 2.1
3	= 2.8
4	= 3.5
5	= 4.2

$$\Rightarrow E = \sum (\hat{y} - y)^2$$

1	= 0.4^2
3	+ 0.4^2
3	+ 0.2^2
2	+ 0.5^2
5	+ 0.8^2

$\Rightarrow E = 3.9 \Rightarrow$ Prediction is 3.9 units away from actual on average.

Note: Pedhazur's method is same as method of least squares which is same as calculus method (this was the derivation)

2) Matrix formulation:

Say we have

$$\left. \begin{aligned} y_0 &= \beta_0 + \beta_1 x_0 \\ y_1 &= \beta_0 + \beta_1 x_1 \\ y_2 &= \beta_0 + \beta_1 x_2 \\ &\vdots \\ y_m &= \beta_0 + \beta_1 x_m \end{aligned} \right\}$$

This can be shown as

$$Y_{m \times 1} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_m \end{bmatrix} \quad X_{m \times 2} = \begin{bmatrix} 1 & x_0 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix}$$

$$\beta_{2 \times 1} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

$$Y = X\beta + \epsilon$$

ϵ → some error
 $\rightarrow$ matrices
 $\rightarrow$ ignore

why not βX ? $\Rightarrow \beta_{2 \times 1} \overset{X}{\underset{\text{No!}}{m \times 2}} \quad X_{m \times 2} \overset{\beta}{\underset{\text{yes!}}{2 \times 1}}$

error $E = (Y - X\beta)^T (Y - X\beta)$

$\downarrow$ differentiate w.r.t. β

$$X^T X \beta = X^T Y \Rightarrow \beta = (X^T X)^{-1} X^T Y \Rightarrow \text{Normal method (direct } \beta)$$

ex: $x \mid y$

1	2
2	3
3	5

$$Y = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \xrightarrow[\text{(calc)}]{\text{inv.}} \begin{bmatrix} 7/3 & -1 \\ -1 & 0.5 \end{bmatrix} = (X^T X)^{-1}$$

$$[X^{-1} = \frac{\text{adj}(X)}{|X|}] \rightarrow \text{transpose of cofactor matrix}$$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 10 \\ 23 \end{bmatrix} = X^T Y$$

$$\Rightarrow X = \begin{bmatrix} 0.333 \\ 1.5 \end{bmatrix}$$

$$\Rightarrow [\beta_0 = 0.333, \beta_1 = 1.5] \Rightarrow \text{eqn} \Rightarrow y = 1.5x + 0.333$$

Multiple Linear Regression (multiple predictors)

$$\Rightarrow \text{say } y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \Rightarrow Y = X\beta + \epsilon$$

$$Y = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_m \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & x_{10} & x_{20} \\ 1 & x_{11} & x_{21} \\ \vdots & \vdots & \vdots \\ 1 & x_{1m} & x_{2m} \end{bmatrix}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

Same solution $\beta = (X^T X)^{-1} X^T Y$

Q) $x = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 3 & 7 \\ 1 & 4 & 8 \end{bmatrix}$, $y = \begin{bmatrix} 20 \\ 25 \\ 32 \end{bmatrix} \rightarrow$ try solving

$\Rightarrow$ Polynomial (quadratic):

trick: say quadratic

$\Rightarrow y = \beta_0 + \beta_1 x + \beta_2 x^2 \rightarrow 3 \text{ variables} \rightarrow 3 \text{ equations}$

eqn ① $\Rightarrow \sum y = \sum \beta_0 + \beta_1 \sum x + \beta_2 \sum x^2 \quad \text{--- (1)}$

apply $\sum$ $\Rightarrow \sum xy = \beta_0 \sum x + \beta_1 \sum x^2 + \beta_2 \sum x^3 \quad \text{--- (2)}$

multiply by x again $\Rightarrow \sum x^2 y = \beta_0 \sum x^2 + \beta_1 \sum x^3 + \beta_2 \sum x^4 \quad \text{--- (3)}$

$\downarrow$ matrix

$$\begin{bmatrix} \sum y \\ \sum xy \\ \sum x^2 y \end{bmatrix} = \begin{bmatrix} n & \sum x & \sum x^2 \\ \sum x & \sum x^2 & \sum x^3 \\ \sum x^2 & \sum x^3 & \sum x^4 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

x	y	$x y$	x^2	x^3	x^4	$x^2 y$
1	2	2	1	1	1	2
2	5	10	4	8	16	20
3	10	30	9	27	81	90
4	17	68	16	64	256	272
$\sum x = 10$	$\sum y = 34$	$\sum xy = 110$	$\sum x^2 = 30$	$\sum x^3 = 100$	$\sum x^4 = 354$	$\sum x^2 y = 384$

$$\Rightarrow \begin{bmatrix} 34 \\ 110 \\ 384 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}^{-1} \begin{bmatrix} 34 \\ 110 \\ 384 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \begin{matrix} \beta_0 = 1 \\ \beta_1 = 0 \\ \beta_2 = 1 \end{matrix}$$

$$\Rightarrow y = 1 + x^2$$

An iterative way to estimate coefficients

③ Gradient descent: (idea) $\beta_0, \beta_1 \rightarrow$ modify $\rightarrow$ check loss

$\hookrightarrow$ Batch: loss function

$$J(\beta_0, \beta_1) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}_i - y_i)^2 \quad : m = \text{no. of elements}$$

$$[\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i]$$

now, like calculus method,

$$\frac{\partial J}{\partial p_0} = \frac{2}{2m} \sum_{i=1}^m (\hat{y}_i - y_i) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)$$

$$\frac{\partial J}{\partial p_1} = \frac{2}{2m} \sum_{i=1}^m (\hat{y}_i - y_i) \cdot x_i = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i) x_i$$

$$p_0 \leftarrow p_0 - \alpha \frac{\partial J}{\partial p_0} \quad \& \quad p_1 \leftarrow p_1 - \alpha \frac{\partial J}{\partial p_1} \quad \left[\begin{array}{l} \alpha = \text{learning} \\ \text{rate} \\ \text{(given)} \end{array} \right]$$

⇒ To check: calculate J & verify

ex:

x	y	$\hat{y} = 0 + 0x$	$\hat{y} - y_i$	$(\hat{y} - y_i) x$
1	2	0	-2	-2
2	3	0	-3	-6
3	6	0	-6	-18

First $m=3$, $\frac{\partial J}{\partial p_0} = \frac{1}{3} (-2 + -3 + -6) = -11/3 = -3.67$
 do multiple $\Rightarrow$ $\frac{\partial J}{\partial p_1} = \frac{1}{3} (-2 + -6 + -18) = -26/3 = -8.67$
 find errors

⇒ $p_0 \leftarrow 0 - 0.1 \times (-3.67) = 0.367$ $p_1 \leftarrow 0 - 0.1 \times (-8.67) = 0.867$
 update

⇒ $\hat{y} = 0.867x + 0.367$, $J_{\text{old}} = \frac{1}{2 \times 3} (4 + 9 + 36) = \frac{49}{6} = 8.167$

⇒

x	y
1	1.234
2	2.101
3	2.968

$J_{\text{now}} = \frac{1}{6} (0.586 + 0.808 + 9.193) = 1.764$ ⇒ error ↓ 😊

error calculation

VS

→ SGD:

loss function: $J(p_0, p_1) = \frac{(\hat{y}_i - y_i)^2}{2}$
 for each observation

$$\frac{\partial J_i}{\partial p_0} = (\hat{y}_i - y_i); \quad \frac{\partial J_i}{\partial p_1} = (\hat{y}_i - y_i) x_i$$

$$p_0 \leftarrow p_0 - \alpha (\hat{y}_i - y_i); \quad p_1 \leftarrow p_1 - \alpha (\hat{y}_i - y_i) x_i$$

Q) $(p_0, p_1) = 0, \alpha = 0.1 \rightarrow$

x	y	$\hat{y} = 0 + 0 \times 1 = 0$
1	2	For (1, 2) $\Rightarrow \hat{y} = 0 + 0 \times 1 = 0$
2	3	$\Rightarrow \frac{dJ}{dP_0} = (0 - 2) = -2$; $\frac{dJ}{dP_1} = (0 - 2) \times 1$
3	6	$\frac{dJ}{dP_1} = -2$

$\Rightarrow P_0 \leftarrow 0 + 0.2$; $P_1 \leftarrow 0 + 0.2$
 $\Rightarrow \hat{y} = 0.2 + 0.2x$

$J_{B4} = \frac{(0-2)^2}{2} = 2$ loss ✓
 $J_{\text{after}} = \frac{(0.2 + 0.2 \times 1 - 2)^2}{2} = 1.28$

For (2, 3) $\Rightarrow \hat{y} = 0.2 + 0.2 \times 2$
 $= 0.2 + 0.4 = 0.6$

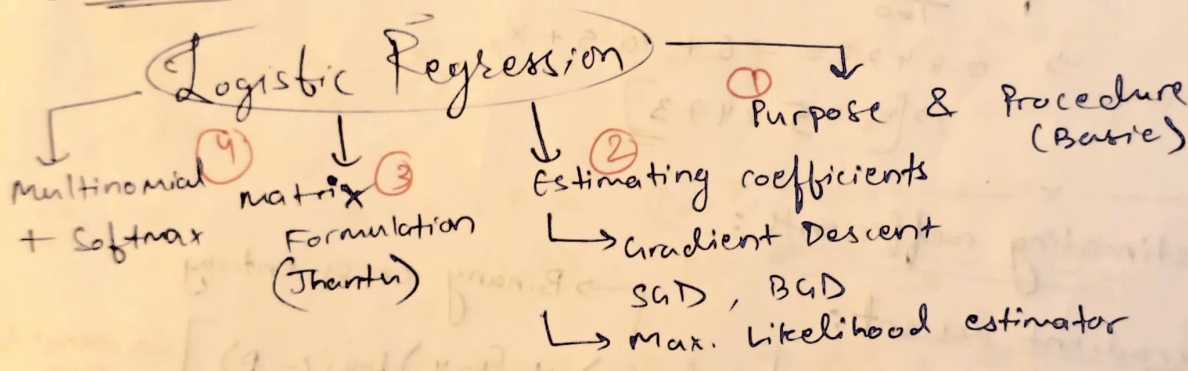
$\frac{dJ}{dP_0} = (0.6 - 3) = -2.4$ $\frac{dJ}{dP_1} = -2.4 \times 2 = -4.8$

$P_0 \leftarrow 0.2 + 0.24 = 0.44$ $P_1 \leftarrow 0.2 + 0.48 = 0.68$

$\Rightarrow \hat{y} = 0.44 + 0.68x$

similarly for (3, 6) ... ✓ solved → For each entry, update the gradient

Section 2:



① say we have a CLASS A & our model says

$$z_A = P_0 + P_1 x_1 + P_2 x_2 + \dots + P_n x_n$$

now A can be 0/1 $\begin{cases} 0: \text{if not A} \\ 1: \text{if A} \end{cases}$

But, z_A doesn't belong strictly b/w 0 & 1

$\Rightarrow$ Apply $\sigma(z) = \frac{1}{1 + e^{-z}} = P(y=1|x)$ (conditional probab.)

Ⓐ $P(y=1|x)$ is read as probability of output belonging to class 1. Such that input is x

$\Rightarrow$ it is obtained by $x \rightarrow z \rightarrow \sigma(z) = P$ 😊

But, $P(y=1|x)$ is not strictly 0/1 $\Rightarrow$ Apply threshold

$\Rightarrow P(y=1|x) > 0.5 ? \Rightarrow 1$
 $\leq 0.5 ? \Rightarrow 0$

Ex: $z = -6 + 0.05x_1 + x_2$ (given) x_1 : hours studied
 x_2 : GPA
 P : probability of A grade

$\Rightarrow a) P(y=A | x_1=40, x_2=3.5)$

English: prob. of earning A grade such that GPA=3.5 hrs & 40 hours studied

$\Rightarrow z = -6 + 0.05 \times 40 + 1 \times 3.5 = -0.5$

$P = \frac{1}{1+e^{0.5}} \Rightarrow P \approx 0.3775 \Rightarrow 37.75\% \text{ chance of A}$
 $\Rightarrow \therefore < 0.5 \Rightarrow \text{prediction: didn't score A}$

$\Rightarrow$ log odds / logits: (for binomial)

$P = \frac{1}{1+e^{-z}}, 1-P = \frac{e^{-z}}{1+e^{-z}} \Rightarrow \ln\left(\frac{P}{1-P}\right) = z$
 $\hookrightarrow$ log odds

Ex: in above Q, min GPA (x_2) for 70% probability of scoring A

if $x_1 = 20$

$\Rightarrow P = \frac{70}{100} = 0.7 \Rightarrow \ln\left(\frac{0.7}{1-0.7}\right) = z = -6 + 0.05 \times 20 + x_2$

$\Rightarrow 0.8473 = -6 + 10.5 + x_2$

$\Rightarrow x_2 = 5.8473$

② Estimating coefficients:

$\hookrightarrow$ Gradient descent:

$\hookrightarrow$ Binary cross entropy

Cost (P, y) = $-[y \log(p) + (1-y) \log(1-p)]$ $\rightarrow$ remember this
predicted probability actual class

$J(P_0, P_1) = -\frac{1}{m} \sum_{i=1}^m [y_i \log(p_i) + (1-y_i) \log(1-p_i)]$

$\Rightarrow$ BGD: $\frac{\partial J}{\partial P_j} = \frac{1}{m} \sum_{i=1}^m (p_i - y_i) x_i$ [same general form as linear regression]
 for $P_0, x_0 = 1$

$P_j \leftarrow P_j - \alpha \frac{\partial J}{\partial P_j}$

Ex: $z = P_0 + P_1 x_1 + P_2 x_2$; $(P_0, P_1 = 0.0)$ $\alpha = 0.1$
 $\sigma(z) = P$ $\textcircled{1}$ make table $\textcircled{2}$ gradient

x_1	x_2	y	P	$\sigma(z) = P$	$P - y$
0	0	0	0	0.5	0.5
0	1	0	0	0.5	0.5
1	0	0	0	0.5	0.5
1	1	1	0	0.5	-0.5

compute gradient
 $\frac{\partial J}{\partial P_0} = \frac{0.5 \times 1 + 0.5 \times 1 + 0.5 \times 1 - 0.5 \times 1}{4}$
 $= 0.25$

$$\frac{dJ}{dp_1} = \frac{0.5 \times (0) + 0.5 \times 1 + 0.5 \times 1 - 0.5 \times 1}{4} = 0$$

$$\frac{dJ}{dp_2} = \frac{0.5(0) + 0.5(1) + 0.5(0) - 0.5(1)}{4} = 0$$

$$\Rightarrow \text{new } p_0 \leftarrow 0 - 0.1 \times 0.25 = -0.025$$

③ update $p_1 \leftarrow 0 - 0.1 \times 0 = 0$ $p_2 \leftarrow 0 - 0.1 \times 0 = 0$

$$\Rightarrow \text{equation: } z = -0.025 + 0x_1 + 0x_2$$

$$\Rightarrow \text{new } p_i \rightarrow z$$

-0.025	0.4937
-0.025	0.4937
-0.025	"
-0.025	"

④ make prediction $J_{bu} = -\frac{1}{4} (0 \log 0.5 + 1 \log 0.5 + 1 \log 0.5 + 1 \log 0.5) = -\frac{\log 0.5}{4} = 0.693$

$$\Rightarrow J_{biter} = -\frac{1}{4} (0 \log (0.4937) + 1 \log (0.5062))$$

$$+ 0 \log (0.4937) + 1 \log (0.5062) + 0 \log (0.4937) + 1 \log (0.5062)$$

$$+ 1 \log (0.4937) + 0 \log (0.5062)$$

$$\Rightarrow -\frac{1}{4} (3 \log (0.5062) + \log (0.4937)) = 0.6870 \text{ less } \textcircled{\smile}$$

$\Rightarrow$ SGD:

$$\frac{dJ}{dp_j} = (p_j - y) x_j \rightarrow p_j \leftarrow p_j - \alpha (p_j - y) x_j$$

$\sim$ SGD of linear regression

ex: x y $z = p_0 + p_1 x : (p_0, p_1) = (0, 0) \quad \alpha = 0.1$

1 0 For (1, 0): $p = \frac{1}{1 + e^{-0}} = 0.5$

2 0

3 1

$$\Rightarrow p - y = 0.5 \rightarrow \frac{dJ}{dp_0} = 0.5 \times 1; \frac{dJ}{dp_1} = 0.5 \times 1$$

$$p_0 \leftarrow 0 - 0.1 \times 0.5 = -0.05$$

$$p_1 \leftarrow 0 - 0.1 \times 0.5 = -0.05$$

$$\Rightarrow z = -0.05 - 0.05 x$$

For (2, 0) $\rightarrow$ same step

$$J_{bu} = !!!$$

where don't report error each iteration but after entire m (3 here) iter

$\Rightarrow$ Max likelihood estimator:

$$L(p) = \prod p_i^{y_i} (1-p_i)^{1-y_i} \Rightarrow \ln(L(p)) = \sum y_i \log(p_i) + (1-y_i) \log(1-p_i)$$

$$\Rightarrow \text{ex: } x_i \quad y_i \quad p(x_i) \Rightarrow L(p) = (1 \times 0.8)^1 \times (1-0.8)^1 \times (1 \times 0.7)^1 \times (1-0.7)^1$$

$$= 0.8 \times 0.2 \times 0.7 \times 0.3 = 0.504$$

$\Rightarrow$ There is 50.4% that model gives correct class output!!

for multinomial: say classes $y = 1$ or 2 or 3

ex: y one hot model o/p $\rightarrow$ will explain later how o/p came

1	(1, 0, 0)	(0.7, 0.2, 0.1)	$\rightarrow (0.7 \times 1 + 0.2 \times 0 + 0.1 \times 0)$
3	(0, 0, 1)	(0.1, 0.3, 0.6)	$\rightarrow (0.1 \times 1 + 0.3 \times 0 + 0.6 \times 0)$

$= 0.42$

③ Matrix formulation:

same as $y = X\beta + \epsilon \rightarrow z = X\beta + \epsilon$ (no ϵ or use case)
just a way to represent

④ Multinomial Logistic Reg?

multi class

say classes A, B, C, each class has own funcⁿ

$$z_A = \beta_{0A} + \beta_{1A}x_1 + \beta_{2A}x_2 \dots$$

$$z_B = \beta_{0B} + \beta_{1B}x_1 + \beta_{2B}x_2 \dots$$

$$z_C = \dots$$

same process

$\Rightarrow$ find z_A, z_B, z_C

$\Rightarrow$ find softmax

$$\frac{e^{z_A}}{\sum_{k=A,B,C} e^{z_k}}$$

$\Rightarrow$ class = $\arg\max(P)$

$\Rightarrow$ ex: $z_A = 0.2 + 0.5x_1 - 0.1x_2$

$z_B = -0.3 + 0.2x_1 + 0.4x_2$

$z_C = 0.1 - 0.4x_1 + 0.3x_2$

say $x_1 = 2, x_2 = 1$

$z_A = 1.1, z_B = 0.5, z_C = -0.4$

$e^{z_A} = 3.004, e^{z_B} = 1.649, e^{z_C} = 0.670$

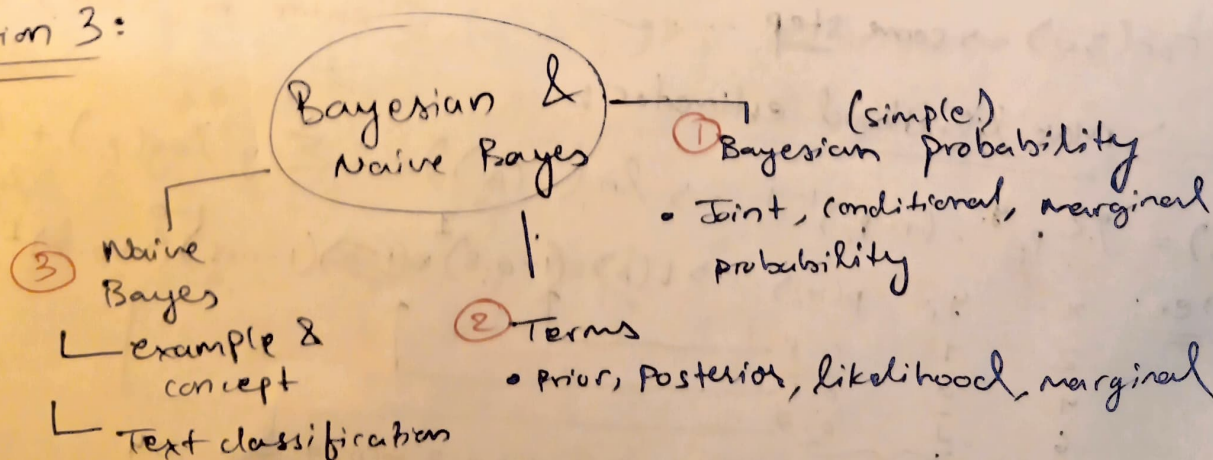
$\Rightarrow P(A) = \frac{3.004}{3.004 + 1.649 + 0.670} = 0.564$

softmax

$P(B) = 0.310, P(C) = 0.126$

$\rightarrow$ largest $\Rightarrow$ predict

Section 3:



Bayesian net, Laplace, Gaussian (Not in syllabus but PYQs)

① Bayesian Prob.:

- Joint probability $P(A, B) = P(A|B) \cdot P(B)$
 - conditional " $P(A|B) \Rightarrow$ P of A occurring : B has occurred
 - Marginal " $P(B) = P(B|A) \cdot P(A) + P(B|\bar{A}) \cdot P(\bar{A})$
- (simple Qn ~ PAO syllabus)

② Terms:

- prior: $P(c) \Rightarrow$ probability of class c prior to any input
 $= \frac{\text{no. of instances of } c}{\text{Total instances}}$
- likelihood: $P(x=v|c) = \text{prob. / likelihood of input being } v \text{ such that class is } c$
 $= \frac{\text{no. of } x=v \text{ where class was } c}{\text{no. of instances of } c}$
- posterior: $P(c|x) = \frac{P(x|c) \cdot P(c)}{P(x)} \Rightarrow \propto \frac{P(x|c) \cdot P(c)}{P(x)}$

$P(x)$ = marginal is same for every class
 note clarity in Qn $\left\{ \begin{array}{l} \text{calculate } S \\ \text{calculate prob.} \end{array} \right.$

③ Naive Bayes

	a	b	c	K = class
1	1	0	1	$\frac{1}{8}$
2	1	1	1	$\frac{1}{8}$
3	1	1	0	0
4	1	0	1	0
5	0	0	0	1
6	0	0	0	1
7	0	0	1	0

calculate:
 $P(K=1 | a=1, b=1, c=0)$

$$\Rightarrow P(K=1) = \frac{4}{8} = 0.5$$

$$P(K=0) = \frac{4}{8} = 0.5$$

$$P(a=1 | K=1) = \frac{2}{4}$$

$$P(b=1 | K=1) = \frac{2}{4}$$

$$P(c=0 | K=1) = \frac{2}{4}$$

$$S_1 \approx P(K=1 | a=1, b=1, c=0)$$

$$= P(K=1) \cdot P(a=1 | K=1) \cdot P(b=1 | K=1) \cdot P(c=0 | K=1)$$

$$= 0.5 \times 0.5 \times 0.5 \times 0.5 = \frac{1}{32}$$

$$S_0 \approx P(K=0 | a=1, b=1, c=0)$$

$$= (\text{same steps}) \Rightarrow 0.5 \times 0.5 \times 0.5 \times 0.25 = \frac{1}{32}$$

$$\Rightarrow P(K=1 | a=1, b=1, c=0) = \frac{1/32}{1/32 + 1/32} = \frac{0.5}{1} = \text{tie!}$$

④ normalize

Naive because all the features assumed independent

④ Text classification:

content	GPU	Nvidia	class = comp. graphics
• NVIDIA is the best GPU	1	1	1
• NVIDIA is giving tough competition to AMD.	0	1	1
• GPU, Ganpat Pandey Univ, is located in Maharashtra	1	0	0

⇒ For a given sentence: "Please buy GPU from our store"
classify as on / not on

⇒ Formulating

$$P(k=1 | GPU=1, Nvidia=0)$$

$$\Rightarrow P(k=1) = \frac{2}{3} \quad P(k=0) = \frac{1}{3}$$

$$P(GPU=1 | k=1) = \frac{1}{2}$$

$$P(NVIDIA=0 | k=1) = 0$$

$$P(GPU=1 | k=0) = \frac{1}{1}$$

$$P(NVIDIA=0 | k=0) = 1$$

$$\Rightarrow S_1 = \frac{2}{3} \times 1 \times 0 = 0$$

$$S_0 = \frac{1}{3} \times 1 \times 1 = \frac{1}{3}$$

$$\Rightarrow P(S_1) = \frac{0}{0+1} = 0 \Rightarrow S_0 \text{ wins}$$

⇒ not computer graphics

Section 4:

KNN — Distances — {manhattan, euclidean, Minkowski} → L_p norm

① Distances:

manhattan = $\sum |x_i - y_i|$

6 (3, 4) → 6 + 8 = 14

8 (9, 12)

euclidean = $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

6 (3, 4) → $\sqrt{36 + 64} = 10$

8 (9, 12)

Minkowski → $(\sum (x_i - x_2)^p)^{1/p} \Rightarrow$ if $p=e \Rightarrow L_p$ norm
if $p=2 \Rightarrow$ euclidean

② ~~calculate~~ calculate distance → sort → choose k points

Q) ex: $k=3$, $y=(6, 8)$

select property
maximum or average

Data:

Student

Point

Res. = Pass

Distance ($y = 6, 8$)

A

(2, 9)

0

$$d(A, y) = \sqrt{4^2 + 1^2} = \sqrt{17}$$

B

(4, 8)

0

$$d(B, y) = \sqrt{4}$$

C

(6, 7)

1

D

(8, 5)

1

E

(3, 6)

1

F

(5, 8)

0

$$\Rightarrow A \rightarrow d(A, y) = 4.123, \underline{d(B, y) = 2}, \underline{d(C, y) = 1}, d(D, y) = 3.606$$

$$d(E, y) = 2.236, \underline{d(F, y) = 1}$$

$$\Rightarrow \{C, F, B\} \text{ closest } k=3 \text{ elements} \Rightarrow \text{Pass} = 1, \text{fail} = 2$$

$\Rightarrow y$ failed!!

Section 5:

$(F, N) \rightarrow N$ means predicted neg.

$\rightarrow$ Your predⁿ was wrong $\Rightarrow$ actually +

Predicted

P

N

Actual

P

TP

FN

N

FP

TN

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \cdot \text{Pr} \cdot \text{Re}}{\text{Pr} + \text{Re}}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

ex:

	P	N
P	3	1
N	2	4

$$P = \frac{3}{5}, R = \frac{3}{4}, A = \frac{7}{10}, F1 = \underline{\underline{\frac{2}{3}}}$$